Behavioral Churn Analytics and Retention Evaluation in Non-Contractual Retail

Bui Chau Anh, Le Ngoc Bich, Tran Thu Hang, Le Nguyen Tue Minh,
Hoang Thi Thanh Nhan, Pham Van Thu

Faculty of Mathematical Economics, National Economics University, Hanoi, Vietnam
Course: Data-driven Marketing

Abstract— This study develops a churn analysis framework for non-contractual retail environments using the Dunnhumby dataset. By applying the Inter-Purchase Time (IPT) method for personalized labeling, we developed machine learning models to identify high-risk customers, achieving a 4.24x lift in the top 5% risk segment. Additionally, through Propensity Score Matching (PSM), the study evaluates the real-world impact of coupon campaigns and reveals that promotions do not always yield positive outcomes. The results provide a practical solution for businesses to optimize retention strategies and allocate marketing resources based on data rather than intuition.

Index Terms— Customer Churn, Retail Analytics, Propensity Score Matching, Retention Marketing

I. Introduction

In modern retail, retaining existing customers is an important source of competitive advantage. Unlike contractual industries such as telecommunications or banking, grocery retailers usually do not observe a formal cancellation event when customers stop purchasing. Therefore, in this context, churn should not be interpreted as contract termination. Instead, it is better understood as an unusual decline in shopping activity relative to each customer's historical purchasing pattern.

This issue is especially relevant for non-contractual businesses, where repeat purchase behavior is the main source of evidence for measuring customer engagement and loyalty. Recent studies show that transaction data can be used to predict future buying behavior and support more efficient marketing resource allocation [11]. In addition, research on churn prediction in non-contractual settings emphasizes that churn identification depends heavily on the label-definition process, because firms cannot directly observe an official customer departure date [13].

Based on these characteristics, this study develops a household-level churn analysis framework using the Dunnhumby dataset. The objective is not to prove that a customer has permanently left the retailer, but to identify households that show abnormal inactivity relative to their own shopping history. The analysis includes four main stages: exploratory data analysis and transaction preprocessing, churn-label construction based on Inter-Purchase Time (IPT), churn-risk prediction modeling, and coupon-campaign impact assessment using Propensity Score Matching (PSM).

II. Exploratory Data Analysis

a. Data Understanding

1. Dataset Overview and Relational Structure

This project uses the Dunnhumby dataset to study customer churn in a grocery retail context. The data are **household-level**, meaning that purchasing behavior is tracked for each household over time rather than only at an aggregate market level. This is important because churn is a customer-level outcome: the objective is to determine whether a household shows signs of disengagement from the retailer.

The dataset is also **relational in structure**. Rather than storing all information in a single table, the data are distributed across several linked tables that describe different aspects of customer activity, including transactions, demographics, product information, campaign participation, coupon records, and marketing exposure. As a result, the first stage of analysis involves not only examining each table separately, but also understanding how these tables connect through shared identifiers such as `household_key` and `PRODUCT_ID`. This relational design provides richer analytical value, but it also requires careful integration before any modeling can be performed.

Within this structure, `transaction_data` is the central table because it captures actual household purchase behavior over time. Each record represents a product-level purchase within a basket and contains the key variables needed to trace customer activity, including purchase day, quantity, sales value, discounts, and store information. For this reason, the transaction table serves as the behavioral backbone of the project, while the remaining tables provide customer, product, and promotional context.

2. Data Components

The dataset consists of eight parquet tables, each representing a different aspect of customer behavior or marketing activity. For clarity, these tables can be grouped into three functional categories. This grouping helps distinguish between direct behavioral evidence, descriptive context, and promotion-related signals, which later support churn labeling, feature construction, and marketing interpretation.

Table 1 summarizes the main data components and their analytical roles in the project.

3. Data Scope and Analytical Sample

The transaction history covers 711 observation days, which provides a sufficiently long horizon for examining repeated shopping behavior and identifying abnormal inactivity patterns. In its raw form, the transaction table contains 2,595,732 records, making it the largest and most behaviorally informative component of the dataset.

To make the main report more informative, several descriptive indicators are retained in the analytical summary. At the raw transaction level, the mean sales value per record is 3.1041, while the median is 2.00. Quantity is highly right-skewed: although the mean quantity is 100.4286, both the median and the 75th percentile are 1, and the maximum reaches 89,638 units. This large gap suggests the presence of extreme operational or bulk-like records and supports the need for later cleaning rules.

After preprocessing and synchronization across related tables, the final analytical sample retains 2,500 valid households and 91,803 valid products. At the household level, the average purchase frequency is 100.4496 shopping baskets, with a median of 73. The average cumulative spending per household is 2,957.3044, while the median is 2,042.2449, indicating that household spending is also right-skewed. Inactivity is relatively low for most households, with a median recency of 6 days, but the distribution is still dispersed because a smaller group of households exhibits much longer inactivity periods. These retained entities form the basis for the subsequent churn-labeling process and later modeling stages. At this stage, frequency, monetary value, and recency are reported only as descriptive indicators of the cleaned household sample. Their analytical roles are formalized later, where recency contributes to churn labeling and frequency and monetary value are further incorporated into feature engineering. Focusing on the cleaned and synchronized sample ensures that later analysis is built on consistent household-product relationships rather than fragmented or anomalous records.

Table 2 reports the core descriptive indicators retained for the main report.

Full schema descriptions and detailed descriptive statistics for individual tables are provided in the Appendix.

b. Data Preprocessing

The preprocessing stage was designed to transform the raw relational retail data into a consistent and analysis-ready customer base for churn modeling. Because the original dataset contains operational records, promotion-related transactions, and extreme purchase quantities that do not always reflect ordinary grocery shopping behavior, a sequence of cleaning and synchronization steps was applied before label construction and feature engineering.

1. Key Harmonization and Table Integration

The first preprocessing task was to standardize the key identifiers used across tables. Variables such as `household_key`, `BASKET_ID`, `PRODUCT_ID`, `STORE_ID`, and `COUPON_UPC` were converted into a consistent string format to avoid join mismatches caused by mixed numeric encodings. This step was essential because the project depends on integrating multiple tables rather than analyzing a single flat file.

After key harmonization, the transaction table was established as the anchor table for later synchronization. Household-level and product-level reference tables were then aligned with the set of valid entities remaining after transaction cleaning. This ensured that the final processed outputs retained only households and products that actually appear in the cleaned behavioral data.

2. Transaction Cleaning Rules

Transaction cleaning focused on removing records that were either behaviorally uninformative or outside the scope of grocery churn analysis. First, all transactions with non-positive `SALES_VALUE` or non-positive `QUANTITY` were excluded. These observations are unlikely to represent regular revenue-generating purchases and may instead reflect returns, zero-value promotional redemptions, or operational adjustments. Applying this rule reduced the transaction table from 2,595,732 records to 2,576,815 records.

Second, department-level filtering was used to remove activities not aligned with the grocery-shopping churn objective. Specifically, transactions associated with `KIOSK-GAS` and `MISC SALES TRAN` were excluded, since these departments represent fuel or miscellaneous operations rather than core supermarket purchasing behavior. Third, extreme quantity values were filtered by removing transactions with `QUANTITY` greater than 150. This threshold was introduced to reduce the influence of bulk-like or clearly anomalous records that could distort behavioral indicators such as spending intensity, purchase frequency, and inter-purchase timing.

After all transaction-cleaning rules were applied, the final cleaned transaction table contained 2,548,770 records. In total, 46,962 rows were removed from the raw transaction table. This corresponds to approximately 1.81% of the original transaction volume, indicating that most of the raw dataset remained usable while the most problematic observations were systematically excluded.

Table 3 summarizes the transaction-cleaning rules and their resulting sample reductions.

3. Final Synchronization and Output Construction

Once transaction cleaning was completed, all related tables were synchronized using the remaining valid household and product identifiers. The product table was restricted to products observed in the cleaned transactions,

TABLE 1: OVERVIEW OF MAIN DATA COMPONENTS

Category	Tables	Definition
Core Behavior	`transaction_data`	Primary behavioral table recording household-product-basket purchase activity. It is used to measure shopping frequency, spending behavior, purchase timing, and inactivity patterns.
Customer and Product Metadata	`hh_demographic`, `product`	Supporting tables that provide descriptive context for observed purchases. The demographic table defines household background characteristics, while the product table defines item attributes such as department, brand, and commodity group.
Promotion and Marketing Exposure	`campaign_table`, `campaign_desc`, `coupon`, `coupon_redempt`, `causal_data`	Promotion-related tables that describe campaign assignment, campaign timing, coupon mapping, coupon redemption behavior, and direct promotional exposure through mechanisms such as displays and mailers.

TABLE 2: CORE DESCRIPTIVE INDICATORS FOR THE MAIN REPORT

Indicator	Value
Observation period	711 days
Initial transaction records	2,595,732
Cleaned transaction records	2,548,770
Removed transaction records	46,962
Valid households	2,500
Valid products	91,803
Mean sales value per transaction	3.1041
Mean household frequency	100.4496
Mean household monetary value	2,957.3044
Mean recency	26.4292
Mean personalized churn threshold	59.2656
Churn rate	12.08%

while household-linked tables such as demographic, campaign, and coupon-redemption records were restricted to valid households. This step ensured coherence across the analytical pipeline and prevented later modeling stages from inheriting orphan entities that no longer had meaningful behavioral observations.

At the assembly stage, the cleaned transactions were enriched with selected product attributes, including department, brand, and commodity description, producing the `transactions_master` table. In parallel, a household-level analytical table was constructed by combining churn-label components, purchase frequency, cumulative spending, and demographic variables. Missing demographic values were imputed as `Unknown` to preserve the full household sample without creating artificial profile information.

The preprocessing pipeline ultimately produced seven processed outputs: `transactions_master`, `customer_base_labeled`, `demographics_imputed`, `campaign_table_clean`, `campaign_desc_clean`, `coupon_clean`, and `coupon_redempt_clean`. These processed tables form the analytical foundation for the subsequent churn-label definition and feature engineering stages.

c. Churn Label Definition

Because the Dunnhumby dataset does not contain an explicit churn flag, churn must be defined from observed purchase inactivity. In a non-contractual grocery setting, a single fixed cutoff is not appropriate because shopping intervals differ substantially across households. This study therefore defines churn relative to each household's own shopping rhythm.

For household i, recency is measured as:

$$\text{Recency}_i = 711 - \text{Last Purchase Day}_i$$

The general personalized threshold is:

$$\text{Threshold}_i = \mu(\text{IPT}_i) + 2\sigma(\text{IPT}_i)$$

For households without enough repeated purchases to estimate a stable household-specific threshold, a global fallback rule is used:

$$\text{Threshold}_i = \mu(\text{IPT}) + 2\sigma(\text{IPT}) = 40.32$$

The churn label is assigned as:

$$\text{is_churn}_i = \begin{cases} 1, & \text{if Recency}_i > \text{Threshold}_i \\ 0, & \text{otherwise} \end{cases}$$

Applying this rule to 2,500 valid households yields 302 churned households and 2,198 retained households, equivalent to a churn rate of 12.08%. This result is reasonable for the present setting: most households remain active, while a smaller subset shows inactivity that is unusually long relative to its own purchase history.

Table 4 summarizes the final labeling rule and the observed label distribution.

This label should be interpreted as behavioral churn, not confirmed customer exit.

TABLE 3: SUMMARY OF TRANSACTION CLEANING RULES

Cleaning Step	Rationale	Result
Remove `SALES_VALUE <= 0` or `QUANTITY <= 0`	Excludes non-standard purchase records such as returns, zero-value redemptions, or operational entries that do not reflect normal customer demand.	2,576,815 records retained
Exclude `KIOSK-GAS` and `MISC SALES TRAN`	Restricts the analysis to relevant grocery-shopping behavior rather than fuel or miscellaneous operational transactions.	Non-grocery departments removed
Filter `QUANTITY > 150`	Limits the impact of extreme or bulk-like purchases that would distort household-level behavioral measures.	2,548,770 records retained

TABLE 4: SUMMARY OF CHURN LABEL CONSTRUCTION

Component	Definition	Observed Result
Behavioral unit	Household-level shopping history based on unique purchase days	2,500 valid households
Personalized threshold	$\mu(\text{IPT}_i) + 2\sigma(\text{IPT}_i)$	Mean threshold = 59.2656 days
Fallback threshold	Global mean IPT + 2 global standard deviations	40.32 days
Recency measure	$711 - \text{Last Purchase Day}_i$	Mean recency = 26.4292 days
Final label distribution	Household classified as churned if recency exceeds threshold	302 churned, 2,198 retained

d. Data Visualization & Key Insights

1. Overall Churn Rate

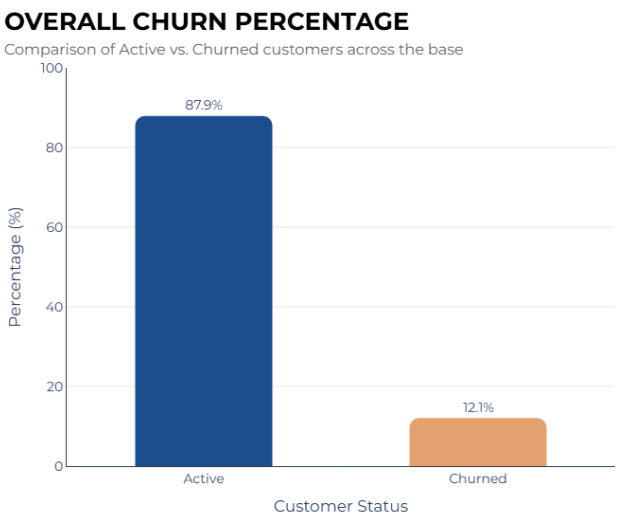

Fig. 1: Overall Churn Rate

Figure 1 visualizes the final label distribution: 2,198 active households (87.9%) and 302 churned households (12.1%). This class imbalance is moderate but still important for later model training and evaluation.

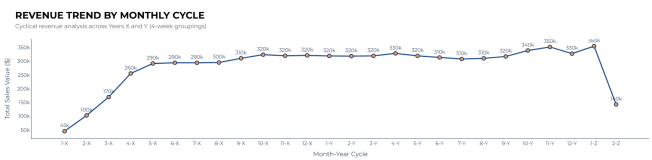

Fig. 2: Sales trend

- **Growth Phase (1-X to 5-X):** Sales surge rapidly from **$45k** to **$290k**, indicating an initial expansion of the active customer base.
- **Maturity Phase (6-X to 1-Z):** Revenue enters a steady state, fluctuating consistently between **$310k** and **$350k**. This stability reflects the reliable spending habits of the loyal core.
- **Data Truncation (2-Z):** The sharp drop to **$140k** in the final cycle is an artifact of the **day 711 cut-off**, where the last month's data is incomplete.

2. Demographics insights

A critical observation regarding the data quality is that while the dataset includes 2,500 customers in total, only about 801 households have demographic information available. This significant data gap is, in itself, a potential churn signal, as customers who are less engaged or unwilling to provide personal information may have lower psychological switching costs and a higher propensity to disengage from the retailer.

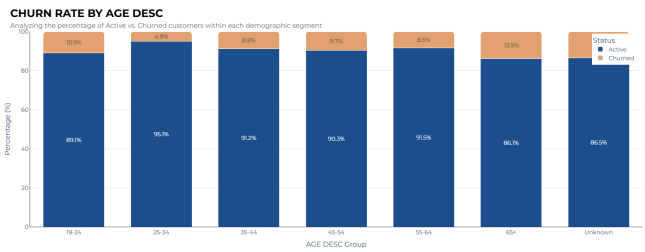

Fig. 3: Age

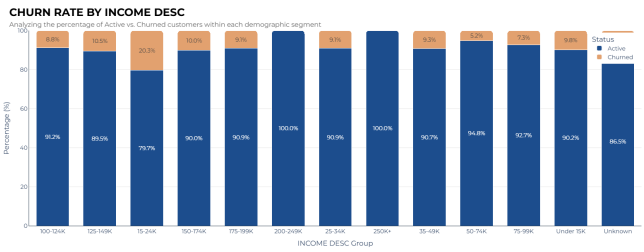

Fig. 4: Income

Churn is disproportionately high among the 65+ age group (13.9%) and low-income households (15–24K), where the churn rate peaks at 20.3%—nearly double the population average. This suggests that the retailer is struggling to retain elderly shoppers and price-sensitive, low-income households.

Stability increases with family size. Larger households (5+ members) exhibit the lowest churn rate at 3.7%. Conversely, single-person households or single-parent units show a much higher risk, with churn rates reaching 12.8%. This suggests that the recurring needs of larger families create a stable shopping routine that naturally anchors them to the retailer.

3. Shopping Rhythm and Behavioral Divergence

The longitudinal shopping rhythm, defined as the average monthly visit frequency per household, reveals a distinct behavioral divergence between active and churned cohorts. As shown in Figure 5, both groups follow a similar trajectory during the initial months, indicating that churned households are not inherently low-engagement customers during their onboarding period.

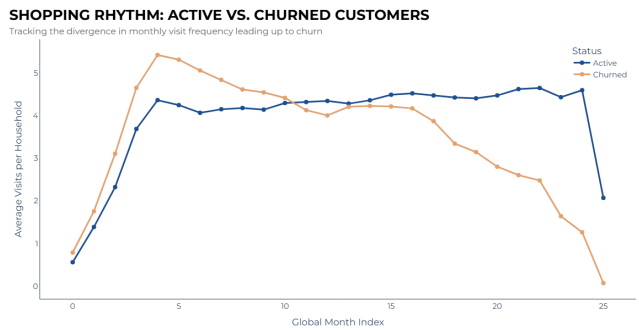

Fig. 5: Divergence in monthly shopping frequency

The analysis identifies three critical observations regarding this divergence:

- **Trial intensity:** Churned households initially exhibit a higher visit frequency than active households, peaking at over 5 visits per month. This confirms that these customers are active trial shoppers who engage significantly with the retailer before their eventual decline.
- **The predictive window:** The two trajectories intersect around the 10th month, marking a decisive shift. While active households sustain a stable rhythm of approximately 4.5 visits per month, the churned group enters a phase of gradual behavioral erosion. This steady decline in visit frequency serves as a primary leading indicator of attrition.
- **The point of no return:** Beyond the 16th month, the churn cohort experiences an accelerated collapse in visit frequency. By the time a household's monthly visits drop below the 2-visit threshold, the customer has likely established shopping routines with competitors, rendering the loss nearly irreversible.

From a strategic perspective, the predictive window identified between months 10 and 13 is the optimal period for marketing intervention. Rather than targeting households that have already ceased purchasing, predictive efforts should focus on identifying the negative slope in visit frequency. This shift in engagement momentum provides the foundational signal for the churn prediction model developed in the subsequent section.

III. Modeling

a. Feature Engineering

Based on the findings from the exploratory data analysis, the feature engineering process is designed to transform raw transactional behavior into highly predictive indicators. To prevent data leakage and accurately simulate a real-world predictive environment, a static time threshold, referred to as the *Cut-off Day*, is established. Specifically, all features are calculated exclusively from transactions occurring before the Cut-off Day, defined as:

$$\text{Cut-off Day} = \text{MAX_DAY} - 60$$

The 60-day holdout horizon is used to separate pre-cutoff feature construction from post-cutoff behavioral assessment and short-horizon value estimation. The churn label itself remains the personalized IPT-based abnormal-inactivity label evaluated at the full observation endpoint.

The engineered and optimized variables are categorized into four core groups. While the complete feature engineering pipeline generates a comprehensive set of 27 variables covering various granular aspects of user behavior, the following sections highlight the most primary and predictive indicators developed for this study.

1. Core Behavioral Features

Extended RFM. Frequency and total spend, or Monetary, are calculated to reflect short-term customer engagement. The Recency variable is capped at 90 days to reduce noise from seasonal absences. In addition, the *Inactive Days Ratio* is normalized based on the customer's lifetime, creating a fairer scale for comparison between newly acquired and long-term customers.

Behavioral Stability: Inter-Purchase Time. Instead of relying only on the average number of days between two purchases, denoted as IPT Mean, the model also extracts the coefficient of variation:

$$\text{IPT_CV} = \frac{\text{IPT_std}}{\text{IPT_mean}}$$

This variable plays an important role in identifying unstable purchasing behavior. A high IPT_CV indicates erratic shopping habits, which may serve as an early warning signal for customers at risk of churning.

2. Momentum and Marketing Features

Brand Affinity. The *Private Brand Ratio* is calculated to measure the proportion of private brand purchases. Laplace smoothing is applied to reduce the influence of outliers among customers with limited transaction history.

Rolling Frequency Trend. Customer churn rarely happens abruptly. To capture the gradual decline in shopping activity, a first-degree polynomial fit is applied to the number of shopping trips over the last three months before the Cut-off Day, denoted as M1, M2, and M3. The resulting frequency slope is used as a momentum indicator. A negative slope provides statistical evidence of declining shopping demand.

Causal Responsiveness. Marketing responsiveness features are integrated from the causal data. Specifically, the model measures the purchase rate in the presence of displays or direct mailers, allowing it to capture customer sensitivity to promotional efforts.

3. Foundation for Causal Inference

To facilitate the causal inference phase in the experimental design section, customers are categorized based on their promotional interaction frequency, measured by *promo_usage_ratio*. Customers whose promo usage ratio exceeds the population mean are assigned to the treatment group:

$$\text{is_treated} = \begin{cases} 1, & \text{if promo_usage_ratio} > \overline{\text{promo_usage_ratio}} \\ 0, & \text{otherwise} \end{cases}$$

Customers in the treatment group are interpreted as *Heavy Promo Users*, while the remaining customers serve as the control group. This classification provides the foundational schema for the subsequent Propensity Score Matching approach.

4. Multicollinearity Audit

Beyond the primary metrics detailed above, the engineering process also extracts numerous auxiliary variables, such as campaign fatigue indicators and demographic proxies, culminating in an initial feature space of 27 variables. To meet the assumptions of different algorithm families and ensure robustness, both the correlation matrix and the Variance Inflation Factor (VIF) are rigorously evaluated.

VIF quantifies the severity of multicollinearity by measuring how much the variance of an estimated regression coefficient increases when predictors are correlated. In practice, a VIF value exceeding 5 is commonly considered a signal of problematic collinearity that may distort model interpretability [8].

Consequently, the final output data is split into two independent versions tailored for specific modeling approaches:

- **Version A: Tree-based Models.** This version retains the complete set of 27 features to fully leverage the capability of algorithms such as Random Forest and XGBoost in capturing complex, non-linear relationships without being hindered by collinearity.
- **Version B: Linear Models.** This version omits structurally redundant variables that cause perfect multicollinearity (e.g., dropping IPT_mean and IPT_std in favor of IPT_CV). This refinement process results in a strictly independent set of 24 features, ensuring the mathematical stability and interpretability of coefficients in the Logistic Regression model.

b. Model Development

1. Prediction Objective and Modeling Sample

The objective of the predictive modeling stage is to estimate whether a household is likely to be classified as churned under the behavioral churn definition developed earlier. In this project, churn is not treated as a contractual cancellation event. Instead, it represents abnormal inactivity relative to a household's own historical shopping rhythm. Therefore, the model should be interpreted as a churn-risk screening system rather than a definitive detector of permanent customer exit.

To simulate a realistic predictive setting, the model uses behavioral, shopping-rhythm, and marketing-related features observed before the cut-off day. The cut-off day is defined as day 651, corresponding to 60 days before the end of the 711-day observation horizon. This prevents post-cutoff behavior from entering the predictor set. The churn target remains the personalized IPT-based abnormal-inactivity label evaluated at the full observation endpoint, rather than a simple fixed rule of no purchase in the next 60 days.

After feature construction, the final modeling matrix contains 2,499 households. Among these households, 302 are labeled as churned and 2,197 are labeled as non-churned, producing a churn rate of approximately 12.08%. This confirms a clear class imbalance problem: the minority class is the group of marketing interest, while most households remain active or retained.

TABLE 5: MODELING DESIGN SUMMARY

Item	Specification
Unit of analysis	Household
Modeling sample	2,499 households
Positive class	Churn / abnormal inactivity
Positive class count	302 households
Churn rate	12.08%
Feature cut-off	Day 651
Outcome endpoint	Day 711 personalized IPT-based inactivity label
Split design	Stratified train / validation / test, approximately 60% / 20% / 20%
Candidate models	Dummy prior, Logistic Regression, Random Forest, Extra Trees, XGBoost
Imbalance handling	Class weighting / `scale_pos_weight`; no SMOTE on validation or test
Primary ranking metric	PR-AUC / Average Precision
Decision-support metric	F2-score and top-k lift

2. Feature Matrix and Preprocessing

The modeling stage uses two related feature matrices generated during feature engineering. The tree-based feature table contains 27 columns, while the linear-model feature table contains 24 columns after removing structurally redundant variables to reduce multicollinearity. The variable `household_key` is retained only as an identifier and is excluded from model training. The target variable is `churn_flag`.

The predictors can be grouped into five main categories. First, RFM-style behavioral intensity variables such as `Frequency`, `Monetary`, `Basket_Value_Std`, and `Avg_Items_Per_Basket` capture purchase volume and basket-level engagement. Second, recency and inactivity variables such as `Recency_Capped`, `Inactive_Days_Ratio`, `customer_lifetime`, and `active_weeks_ratio` measure recent disengagement and exposure time. Third, shopping-rhythm variables such as `IPT_mean`, `IPT_std`, `IPT_CV`, and `Rolling_Freq_Slope` capture irregularity and declining momentum. Fourth, promotion and campaign variables such as `promo_usage_ratio`, `coupon_dependency`, `Mailer_Responsiveness`, `Display_Responsiveness`, and campaign-count variables measure historical marketing interaction patterns. Finally, contextual variables such as `Private_Brand_Ratio`, `has_demographic_info`, and `Primary_Store_ID` capture store and household-level context.

For Logistic Regression, numeric predictors are standardized and `Primary_Store_ID` is one-hot encoded. This is necessary because the linear model is sensitive to feature scale and categorical representation. For tree-based models, numeric features are not standardized, because tree splits are not affected by monotonic rescaling. The tree-based feature version retains the fuller feature set, while the linear version removes highly redundant variables to improve coefficient stability.

3. Candidate Models and Class Imbalance Handling

Five churn-classification models are benchmarked. A Dummy prior model is included as a naive baseline. Logistic Regression with balanced class weights is used as an interpretable linear benchmark. Random Forest and Extra Trees are included as nonlinear ensemble models that can capture interactions among behavioral variables. XGBoost is included as a gradient boosting benchmark and uses `scale_pos_weight` to account for the minority churn class.

Because churned households account for only about 12.08% of the sample, accuracy is not used as the main evaluation criterion. A model that predicts most households as non-churn could appear accurate while failing to identify the minority class. The modeling pipeline therefore emphasizes PR-AUC, F2-score, recall, precision, Brier score, calibration quality, and top-k lift. Precision-recall analysis is especially relevant for imbalanced classification because precision directly measures the proportion of true churners among households predicted as high risk [19].

Class imbalance is handled through model-level weighting rather than synthetic resampling in the final pipeline. Logistic Regression uses `class_weight = balanced`, Random Forest uses `class_weight = balanced_subsample`, Extra Trees uses `class_weight = balanced`, and XGBoost uses `scale_pos_weight`. SMOTETomek is disabled in the final model configuration. This choice is important because the downstream business logic uses predicted probabilities directly. Synthetic resampling can distort the relationship between predicted probability and the true churn rate if not handled carefully. Validation and test sets are not resampled.

4. Model Selection, Calibration, and Thresholding

Candidate models are compared using stratified train, validation, and test splits. The training set is used for fitting and hyperparameter tuning, the validation set is used for model selection, calibration-method selection, and threshold tuning, and the test set is held out for final reporting.

The final reporting model is Logistic Regression with balanced class weights and isotonic calibration. This does not mean that Logistic Regression is universally superior to every tree-based alternative. In fact, several tree-based models show competitive raw ranking performance. The reporting champion is selected because it provides the preferred trade-off under the validation-calibration protocol: calibrated probability quality, recall-oriented decision support, and acceptable ranking performance.

Calibration is necessary because raw model probabilities are not automatically reliable, especially when class weights are used. Prior work on probability estimation shows that strong ranking performance does not guarantee well-calibrated posterior probabilities, so calibration is important when predicted probabilities are used for downstream decision-making [14]. For example, the raw Logistic Regression model produces a mean predicted churn probability of 45.41% on the test set, while the actual churn rate is only 12.00%. This indicates severe probability inflation. After isotonic calibration, the mean predicted probability becomes 12.13%, closely matching the actual test churn rate of 12.00%. The calibration gap is therefore reduced to approximately 0.13 percentage points.

The calibration method is selected using a validation-based rule: calibration methods whose validation Brier score is within 0.015 of the best-Brier method are considered eligible, and the method with the highest validation PR-AUC among eligible candidates is selected. Under this rule, isotonic calibration is selected for the Logistic Regression champion.

The final classification threshold on calibrated probability is 0.07. This low threshold should not be interpreted as a universal churn cutoff. It is a screening-oriented threshold selected on the validation set to support recall-heavy churn detection. In a retention-marketing context, missing likely churners can be more costly than flagging some extra households for low-cost screening. However, this threshold is not sufficient by itself to justify expensive interventions.

5. Evaluation Metrics

The model is evaluated using several complementary metrics. PR-AUC is used as the primary ranking metric because the positive class is rare and the marketing objective

is to prioritize a limited set of high-risk households. ROC-AUC is reported as a general discrimination measure but is not treated as the primary criterion. F2-score is used as the threshold-based metric because it weights recall more heavily than precision. Precision and recall are both reported to show the trade-off between false positives and missed churners. Brier score, mean predicted probability, and calibration gap are used to evaluate probability reliability. Finally, top-k precision and lift are used to evaluate whether the model can identify a concentrated high-risk customer subset for downstream marketing analysis.

c. Model Interpretation

1. Overall Predictive Performance

Table 6 compares the final test-set performance of the candidate models and the calibrated champion. The Dummy prior baseline achieves a PR-AUC of 0.1200, which corresponds to the approximate base churn rate. All trained models improve over this baseline in terms of PR-AUC, indicating that the engineered behavioral and campaign-related variables contain useful churn-risk signal.

TABLE 6: TEST-SET MODEL PERFORMANCE COMPARISON

Model	PR-AUC	F2	Prec.	Recall	Brier
Dummy prior	0.1200	0.4054	0.1200	1.0000	0.1056
LR raw	0.3421	0.4574	0.1826	0.7333	0.2052
XGBoost	0.3643	0.4602	0.2284	0.6167	0.1697
Extra Trees	0.3682	0.4338	0.1919	0.6333	0.1105
Random Forest	0.3654	0.4685	0.1651	0.8667	0.0998
LR + isotonic	0.3185	0.4721	0.1947	0.7333	0.0933

Note: Champion selection prioritized calibrated probability quality and recall-oriented decision support, not raw PR-AUC alone. The selected calibrated threshold is 0.07.

The tree-based models produce stronger raw test PR-AUC than the calibrated Logistic Regression champion. However, their raw probabilities are not selected as the final business-facing scores. The reporting champion is selected because calibrated probability quality is essential for downstream expected-profit calculations and risk-prioritization decisions. The calibrated Logistic Regression model achieves the strongest probability-quality trade-off among the final reporting outputs and produces a mean predicted churn probability close to the actual churn rate.

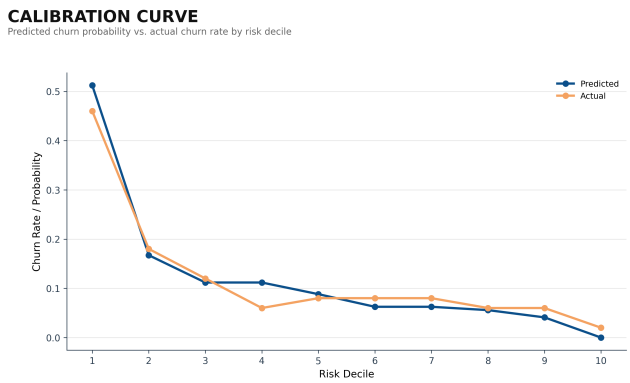

Fig. 6: Calibration curve of the selected churn model.

The calibrated model produces a mean predicted churn probability of 12.13%, close to the actual test churn rate of 12.00%. The final test confusion matrix for the calibrated champion is shown in Table 7. The model detects 44 out of 60 churned households in the test set, corresponding to a recall of 73.33%. However, it also flags 182 non-churn households as churn-risk, yielding a precision of 19.47%. This confirms that the model is not appropriate as an automatic trigger for expensive retention incentives. Its more appropriate use is screening and prioritization: identifying a smaller group of households with elevated churn risk for further analysis, low-cost outreach, or controlled experimentation.

TABLE 7: CHAMPION CONFUSION MATRIX ON TEST SET

	Pred. non-churn	Pred. churn
Actual non-churn	258	182
Actual churn	16	44

2. Churn-Risk Ranking and Top-k Targeting

Although the binary threshold produces many false positives, the model performs better when used as a ranking system. This is more aligned with the marketing objective, where a firm rarely targets every customer above a fixed probability threshold. Instead, the firm usually selects the highest-risk subset subject to budget and operational constraints.

TABLE 8: TOP-K CHURN-RISK TARGETING PERFORMANCE

Group	Cust.	Prec. at k	Lift	Recall at k
Top 5%	125	51.2%	4.24x	21.2%
Top 10%	250	42.0%	3.48x	34.8%
Top 20%	500	32.8%	2.71x	54.3%

The top 5% of households ranked by calibrated churn probability has a churn concentration of 51.2%, which is approximately 4.24 times the baseline churn rate. The top 10% captures 105 churned households with 42.0% precision, while the top 20% captures 164 churned households, or 54.3% of all churned households, with 32.8% precision. These results show a clear trade-off between targeting efficiency and churner coverage. Smaller top-risk groups provide higher precision and are more suitable for limited campaign budgets, while larger groups capture more churners at the cost of more false positives.

This ranking behavior supports the use of the model as a churn-risk prioritization layer. The calibrated scores can help identify a concentrated high-risk subset for downstream causal analysis and marketing experimentation. However, the predictive model itself does not estimate causal treatment effects.

3. Feature Contribution and Behavioral Interpretation

Feature-contribution analysis is based on SHAP values, which provide additive feature-attribution

explanations for model predictions [10]. As shown in Figure 7, recency and inactivity-related variables dominate churn-risk prediction. The most important global SHAP features include `Recency_Capped`, `Inactive_Days_Ratio`, `Camp_Count_TypeA`, `Campaigns_Last_30D`, `Mailer_Responsiveness`, `Rolling_Freq_Slope`, `active_weeks_ratio`, `IPT_std`, `Days_Since_Last_Camp`, and `Avg_Items_Per_Basket`.

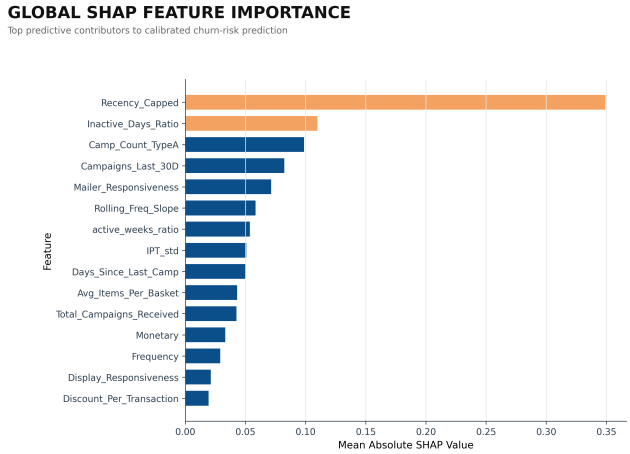

Fig. 7: Global SHAP feature importance for churn-risk prediction.

Recency and inactivity-related variables dominate the prediction, followed by campaign-contact and purchase-rhythm features.The strongest drivers are consistent with the churn definition and with the exploratory analysis. `Recency_Capped` and `Inactive_Days_Ratio` directly reflect recent disengagement and prolonged inactivity. Shopping-rhythm variables such as `Rolling_Freq_Slope` and `IPT_std` capture declining or unstable purchase routines, which are plausible early warning signals in grocery retail. Basket-level features such as `Avg_Items_Per_Basket`, `Monetary`, `Frequency`, and `Basket_Value_Std` represent spending intensity and engagement depth.

Campaign-related variables also contribute to the model's predictions. Variables such as `Camp_Count_TypeA`, `Campaigns_Last_30D`, `Mailer_Responsiveness`, and `Days_Since_Last_Camp` help the model distinguish households with different histories of marketing contact and promotional response. However, these variables must be interpreted carefully. SHAP values explain contributions to model predictions; they do not prove that campaign exposure causes churn or retention. The correct interpretation is predictive association, not causal effect.

4. Short-Horizon Value and Expected-Profit Interpretation

In addition to churn-risk prediction, the model estimates short-horizon discounted customer value to support business prioritization. This should not be described as full lifetime CLV. The value layer estimates discounted 60-day customer value conditional on future activity. The selected conditional value model is a Random Forest Regressor trained on the target $\log(1 + \textit{discounted_future_revenue_60d} \mid \textit{active})$. On the test set, this model achieves an RMSE on the log scale of 0.8851, an MAE on the log scale of 0.6417, an R^2 of 0.4973, and a revenue-scale MAE of 143.77.

Expected incremental profit is calculated as:

$$\begin{aligned}\mathrm{EIP}_i = {} & p(\mathrm{churn})_i \times \mathrm{save_rate} \\ & \times \mathrm{value_if_active}_i \times \mathrm{gross_margin} - \mathrm{treatment_cost}.\end{aligned}$$

This formula combines calibrated churn risk, assumed treatment effectiveness, predicted short-horizon value, margin, and treatment cost. The formula is scenario-based because save rate, gross margin, and treatment cost are assumptions rather than experimentally validated treatment effects.

TABLE 9: EXPECTED-PROFIT SCENARIO SUMMARY

Scenario	Margin	Save	Cost	Prof. cust.
Conservative	0.20	0.03	5.0	0
Base	0.25	0.05	5.0	0
Optimistic	0.30	0.08	3.0	6

Under the base scenario, no household has positive expected incremental profit. The conservative scenario also produces zero positive-profit customers, while the optimistic scenario identifies only six positive-profit households. Therefore, the model does not justify broad paid retention campaigns under the reported assumptions. This is an important business finding rather than a modeling failure: the economics of paid treatment depend not only on churn risk, but also on value, margin, treatment cost, and true causal lift.

5. Business Use and Limitations

The pipeline is most appropriate as a churn-risk screening and prioritization system. It improves the concentration of churned households in the top-ranked risk groups and produces calibrated probabilities that are more realistic than raw weighted-model outputs. This makes the model useful for identifying customers who may deserve closer monitoring, low-cost outreach, or inclusion in downstream causal analysis.

However, several limitations must be acknowledged. First, the validation is internal to the same dataset; no external validation is performed. Second, the churn label is behavioral abnormal inactivity, not an observed cancellation or permanent retailer defection. Third, the binary classification threshold has low precision, and the model produces many false positives. Fourth, SHAP and feature-importance outputs are predictive explanations only and should not be interpreted causally. Fifth, the base expected-profit scenario has zero positive-profit customers, so the model should not be used to automatically trigger expensive incentives.

The correct business interpretation is therefore balanced. The model can identify a high-risk subset whose churn concentration is substantially above the population baseline. It can support risk stratification and prioritization for the subsequent causal inference framework. However, it does not prove that any promotion or coupon treatment will reduce churn. Treatment effectiveness and profit impact must be evaluated separately through the causal inference stage before any broad retention campaign is recommended.

IV. Causal Inference Framework: Assessing the Impact of Coupon Campaigns on Customer Churn

a. Study Design and Variables

The analysis draws on transaction records from 2,499 households. To assess the impact of marketing interventions, we established a strict temporal separation: observations prior to day 651 were used to construct behavioural features, while the subsequent 60-day window (days 652–711) defined the binary outcome variable. Crucially, this outcome variable (*churn_flag*) is defined differently from the personalized IPT-based label used in Sections II and III. While the IPT method identifies individual long-term risk, causal inference requires a standardized observation window to ensure that the treatment effect is measured over a uniform period for both groups, allowing for a direct comparison of purchase probabilities within the immediate marketing cycle. This operationalises churn as sustained behavioural inactivity, consistent with the non-contractual retail setting [5]. The binary treatment indicator (*is_treated* = 1) identifies households that exhibited above-average promotional engagement prior to day 651; this assignment is historical and non-randomised.

Because coupons were historically directed at high-value customers–who are intrinsically less likely to churn–a naive comparison of churn rates across groups would suffer from *selection bias* [6]. To mitigate this, we employed Propensity Score Matching (PSM), which constructs a comparable control group without requiring a randomised experiment [18].

Fifteen behavioural covariates were used for matching, covering purchase frequency, recency, monetary value, inter-purchase time dynamics, and engagement patterns (see Table 10). A predicted churn-risk decile (*risk_decile*), derived from the classification model in the preceding phase, served as a stratification variable.

Table 10: Behavioural Covariates Used for Propensity Score Estimation

Variable	Description
Frequency	Purchase count over the observation window
Monetary	Total expenditure
Basket_Value_Std	Std. dev. of basket value
Customer_Lifetime	Days between first and last purchase
Recency_Capped	Days since most recent purchase (winsorised)
Inactive_Days_Ratio	Proportion of days with no purchase
Active_Weeks_Ratio	Proportion of weeks with at least one purchase
Avg_Items_Per_Basket	Mean items per transaction
IPT_Mean	Mean inter-purchase time
IPT_Std	Std. dev. of inter-purchase time
IPT_CV	Coefficient of variation of inter-purchase time
Private_Brand_Ratio	Share of spend on private-label products
Rolling_Freq_Slope	Temporal trend in purchase frequency
Days_Since_Last_Camp	Days since last marketing contact
Has_Demographic_Info	Demographic data available (binary)

b. Rationale for Propensity Score Matching

Estimating the causal effect of coupon distribution is complicated by the absence of a randomised experiment. In observational settings, treated and control units often differ systematically on pre-treatment characteristics–a problem known as *selection bias* [6]. PSM addresses this by conditioning on the *propensity score*: the probability of receiving treatment given a set of observed covariates [18]. Under the assumption of strong ignorability–that treatment assignment is independent of potential outcomes conditional on observed covariates–matched units can be treated as approximately exchangeable, enabling unbiased estimation of the ATE [18, 1].

Compared to regression-based adjustment, PSM offers a transparent balancing procedure that does not require specifying a functional form for the outcome model [4]. It is particularly suited to the present context, where (i) the treated group is a non-random subset of the customer base, (ii) no randomised control trial data are available, and (iii) a rich set of behavioural covariates allows for credible propensity score estimation.

c. Segment Stratification

Prior to matching, customers were stratified into two segments based on their risk decile, consistent with RFM theory [7] and uplift modelling principles [16]:

- **Target segment** (deciles 3, 5, 6, 7): moderate-risk customers considered *activatable*–neither too likely nor too unlikely to churn for an intervention to matter.
- **Non-Target segment** (deciles 1, 2, 4, 8, 9, 10): customers at extreme risk levels or with stable loyalty patterns. High-loyalty customers in particular may exhibit a *sleeping dogs* effect, whereby unsolicited promotions accelerate rather than prevent churn [9].

Matching was conducted separately within each segment to improve covariate balance [20].

d. Propensity Score Matching Procedure

The propensity score–defined as $e(\mathbf{x}) = P(W = 1 \mid \mathbf{X} = \mathbf{x})$, the conditional probability of *exhibiting high promotional engagement* (treatment) given the observed behavioural covariates [18]–was estimated using a logistic regression model fitted on the 15 covariates listed in Table 10. Note that the propensity score models the likelihood of treatment assignment, not the likelihood of churn; conflating the two would invalidate the balancing property on which PSM relies [1]. The matching procedure followed three steps:

1. **Common support restriction.** Ninety-one observations whose propensity scores fell outside the overlapping support region were excluded to ensure comparability [4].
2. **Nearest-neighbour 1:1 matching with caliper.** Each treated unit was matched to the nearest untreated unit within a caliper of 0.05 on the propensity score scale. To eliminate potential ordering bias, treated units were randomly shuffled prior to matching. The

procedure was applied independently within each segment, discarding units with no admissible match (14 in the Target segment; 81 in the Non-Target segment), yielding 283 and 396 matched pairs, respectively.

3. **Balance assessment.** Post-matching covariate balance was evaluated using the Standardised Mean Difference (SMD):

$$\text{SMD} = \frac{\bar{X}_{\text{treated}} - \bar{X}_{\text{control}}}{\sqrt{\frac{s^2_{\text{treated}} + s^2_{\text{control}}}{2}}} \tag{1}$$

An SMD below 0.10 is conventionally considered indicative of adequate balance [1]. As reported in Table 11, the Non-Target segment achieved balance across all 15 covariates, while three covariates in the Target segment remained above threshold (*Inactive_Days_Ratio* $= 0.116$; *Recency_Capped* $= 0.109$; *IPT_Std* $= 0.106$). Consequently, Average Treatment Effect (ATE) estimation was restricted to the Non-Target segment.

TABLE 11: POST-MATCHING BALANCE ASSESSMENT BY SEGMENT

Segment	Pairs	Max SMD	Balanced
Non-Target (D1,2,4,8,9,10)	396	< 0.10	Yes
Target (D3,5,6,7)	283	0.116	No

e. Treatment Effect Estimation

The ATE is defined as the difference in churn probability between matched control and treated units:

$$\text{ATE} = P(\text{churn} \mid \text{control}) - P(\text{churn} \mid \text{treated}) \tag{2}$$

A positive ATE indicates coupon-induced churn reduction; a negative ATE indicates a counterproductive effect. Statistical significance was assessed using the McNemar test [12], which accounts for the paired structure of matched data and avoids the independence assumption of standard chi-squared tests.

TABLE 12: ATE ESTIMATION RESULTS – NON-TARGET SEGMENT

Metric	Value
Churn rate, control (matched)	8.84%
Churn rate, treated (matched)	13.89%
ATE	−5.05%
Relative lift	+57.14%
Pearson χ^2 p-value	0.033
McNemar p-value	0.034

Results for the Non-Target segment are presented in Table 12. The coupon was associated with a statistically significant *increase* in churn of 5.05 percentage points ($p = 0.034$ under both McNemar and Pearson chi-squared tests), consistent with the sleeping dogs phenomenon [9]. Because the Target segment did not achieve adequate balance, ATE estimation for that segment was not performed; any such estimate would carry residual confounding and should not be interpreted causally. This statistically significant negative effect among Non-Target households reinforces the need for precision in marketing; over-exposure to promotions for customers who are already loyal or highly disengaged (Sleeping Dogs) can be counterproductive.

f. Limitations

Several limitations should be acknowledged. First, PSM controls only for observed confounders; unobserved heterogeneity may still bias the estimates [18]. Second, the failure to achieve balance in the Target segment underscores the sensitivity of PSM to distributional differences within small subgroups; more flexible methods such as inverse probability weighting or doubly robust estimation [2] may be warranted in future work. Third, the segment stratification was partly data-driven, introducing a risk of exploratory bias; conclusions regarding the Target segment remain tentative until confirmed by a randomised controlled trial.

V. CONCLUSION

This study demonstrates that churn in non-contractual retail should be approached as a behavioral signal rather than an absolute state of defection. Utilizing personalized shopping rhythms better reflects the differences between households and provides a logical foundation for churn-risk prediction.

In terms of practical application, the predictive model is primarily valuable for screening and ranking customers, helping businesses identify specific groups for monitoring or intervention testing. However, the results also suggest that high churn risk should not be automatically equated with the need for coupons. Marketing campaigns must be evaluated carefully, as poorly targeted promotions can lead to adverse effects or lack economic efficiency.

Several limitations remain in this research. First, the churn label is based on atypical inactivity, which does not definitively prove permanent customer exit. Second, the model has only been tested on an internal dataset and lacks external validation. Third, Propensity Score Matching (PSM) only accounts for observed confounders, leaving the potential for bias from unrecorded variables.

Future research could expand in three directions: validating the model on new or out-of-sample data, incorporating more robust causal inference methods such as inverse probability weighting or doubly robust estimation, and designing randomized A/B tests to directly measure the impact of retention campaigns.

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APPENDICES

A. Table Inventory

TABLE 13: ROW AND COLUMN INVENTORY OF MAIN TABLES

Table Name	Stage	Role	Rows	Cols
transaction_data	Raw	Core purchase behavior	2,595,732	12
hh_demographic	Raw	Household profile	801	8
product	Raw	Product metadata	92,353	7
campaign_table	Raw	Campaign assignment	7,208	3
campaign_desc	Raw	Campaign schedule	30	4
coupon	Raw	Coupon-product mapping	124,548	3
coupon_redempt	Raw	Coupon redemption behavior	2,318	4
causal_data	Raw	Display and mailer exposure	36,786,524	5
transactions _master	Processed	Cleaned and enriched transaction behavior	2,548,770	15
customer_base _labeled	Processed	Household-level churn modeling base	2,500	16

B. Numeric Descriptive Statistics

TABLE 14: RAW TRANSACTION NUMERIC SUMMARY

Variable	Mean	Std	Min	Q1	Median	Q3	Max
Day	388.76	189.72	1	229	390	553	711
Quantity	100.43	1153.44	0	1	1	1	89638
Sales value	3.10	4.18	0	1.29	2	3.49	840
Retail discount	-0.54	1.25	-180	-0.69	-0.01	0	3.99
Transaction time	1561.59	399.84	0	1308	1613	1843	2359
Week number	56.22	27.10	1	33	56	80	102
Coupon discount	-0.0164	0.2168	-55.93	0	0	0	0
Coupon match discount	-0.0029	0.0397	-7.7	0	0	0	0

TABLE 15: HOUSEHOLD-LEVEL NUMERIC SUMMARY AFTER LABELING

Variable	Mean	Std	Min	Q1	Median	Q3	Max
Mean IPT	17.82	28.97	1.26	5.54	9.54	18.92	589
Std IPT	21.07	32.33	0.60	4.99	10.40	23.19	461.74
Last purchase day	684.57	64.31	54	691	705	709	711
Personalized threshold	59.27	86.68	2.47	15.86	30.83	65.52	1262.98
Recency	26.43	64.31	0	2	6	20	657
Is churn	0.1208	0.3260	0	0	0	0	1
Frequency	100.45	104.76	1	36	73	129	1223
Monetary	2957.30	2970.50	8.17	916.11	2042.24	4024.26	26285.33

C. Selected Categorical Distributions

TABLE 16: SELECTED DEMOGRAPHIC DISTRIBUTION

Variable	Category	Count	Share
Age group	45–54	288	0.3596
Age group	35–44	194	0.2422
Marital status	U	344	0.4295
Marital status	A	340	0.4245
Income	50–74K	192	0.2397
Homeowner status	Homeowner	504	0.6292
Household composition	2 Adults No Kids	255	0.3184
Household size	2	318	0.3970
Kid category	None/Unknown	558	0.6966

TABLE 17: SELECTED PRODUCT DISTRIBUTION

Variable	Category	Count	Share
Department	GROCERY	39,021	0.4225
Department	DRUG GM	31,529	0.3414
Brand	National	78,537	0.8504
Brand	Private	13,816	0.1496
Commodity group	GREETING CARDS/ WRAP/PARTY SPLY	2,785	0.0302
Commodity group	CANDY - PACKAGED	2,475	0.0268
Commodity group	MAKEUP AND TREATMENT	2,467	0.0267